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GLASSBOX: PERSONALIZED CALIBRATION FOR PRIVACY-PRESERVING AND EXPLAINABLE RISK SCORING IN ONLINE EXAM PROCTORING

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BONAFIDE CERTIFICATE

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ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DPDPA	Digital Personal Data Protection Act
FPN	Feature Pyramid Network
FPR / FNR	False Positive Rate / False Negative Rate
FPS	Frames Per Second
GDPR	General Data Protection Regulation
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LR	Logistic Regression
ML	Machine Learning
REST	Representational State Transfer
ROC	Receiver Operating Characteristic
SQL	Structured Query Language
SSD	Single Shot Detector
UI	User Interface
XAI	Explainable Artificial Intelligence

ABSTRACT

Online examinations increasingly rely on automated proctoring, but conventional systems often use fixed behavioral standards and continuous video capture, raising concerns about privacy and unfair risk assessments. This project presents **GlassBox**, a browser-based, privacy-preserving, and explainable risk-scoring system for online examinations.GlassBox extracts gaze and head-pose signals using MediaPipe Face Landmarker and combines them with browser events, including tab visibility changes and window focus loss. A guided calibration process establishes each student's personal behavioral baseline, allowing subsequent observations to be evaluated against individual behavior rather than population-wide thresholds. Raw video is processed locally and never transmitted to the backend; only derived numerical signals, evidence, and feature values are stored and analyzed. The system integrates evidence aggregation, temporal episode detection, feature extraction, and risk classification to assign Low, Medium, or High risk levels with signal-specific explanations. An Express backend and SQLite database support session persistence, while the Teacher Dashboard enables instructors to review flagged sessions and supporting evidence. An interpretable Logistic Regression classifier and a FairFace-based fairness audit support transparent risk assessment and fairness evaluation. GlassBox does not automatically terminate examinations; final decisions remain with human instructors. However, browser-based monitoring cannot reliably detect a second device positioned outside the camera's field of view.

Keywords: *Online exam proctoring, personalized calibration, explainable AI, privacy-preserving monitoring, algorithmic fairness, gaze tracking, logistic regression.*

CHAPTER 1. INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Introduction

Online and remote examinations are now a normal part of education. Because the examiner cannot be physically present, many institutions rely on automated proctoring tools that watch the student through a webcam and the browser. Most of these tools apply one fixed detection standard to every student: the same gaze limit, the same head-movement limit and the same activity thresholds, regardless of how an individual naturally behaves.

This one-size-fits-all approach has two recurring problems that are discussed in the literature. First, face-analysis components can perform unevenly across demographic groups, which may lead to more false alerts for some students. Second, behavior-based rules can penalize students whose normal behavior differs from the assumed norm, for example students who look away while thinking or who need to move during an exam. Tools that continuously record video and audio add a separate privacy concern.

GlassBox is a browser-based proctoring layer designed around a different idea: instead of judging a student against a population norm, it records the student's own baseline at the start of the exam and scores later behavior as deviation from that personal baseline. The processing of the camera image stays inside the browser, the scoring model is deliberately simple and interpretable, and the system only produces evidence for a human instructor — it never penalizes a student automatically.

Problem Statement

Existing automated proctoring systems have the following limitations:

- **Single population-wide standard:** every student is compared with the same fixed thresholds, so natural individual differences are treated as suspicious deviations.
- **Uneven detector performance:** face-analysis components can show confidence gaps across demographic groups, which may translate into unequal false alerts.
- **Dependence on one noisy signal:** many tools rely mainly on gaze, which is easily disturbed by lighting, posture and camera position.
- **Privacy exposure:** continuous video and audio capture is sent to or stored by a third party.
- **Opaque decisions:** a score or alert is given without showing which behavior produced it, so a student has little chance to contest it.

- **Automatic penalties:** some systems act on the alert without a human reviewing the evidence.

There is therefore a need for a browser-based, privacy-preserving proctoring layer that adapts to the individual student, explains its output and keeps the instructor in control of every decision.

Objectives of the Project

The main objectives of this project are:

1. **Real-time detection:** track gaze direction, head pose, tab switching and window focus loss directly in the browser, with no server-side video processing.
2. **Personalized calibration:** record each student's own behavioral baseline in a guided sequence at the start of the exam instead of applying one fixed rule to everyone.
3. **Temporal, multimodal evidence:** group signals into behavioral episodes and fuse them, rather than flagging isolated events.
4. **Transparent risk output:** produce an explainable Low / Medium / High risk band that shows which signals contributed.
5. **Accommodation-aware scoring:** give accommodated students individually shifted thresholds within the same scoring formula, not a separate system.
6. **Fairness audit:** compare the detection confidence of MediaPipe and an independently trained detector (RetinaFace) across demographic groups on the FairFace dataset, offline.
7. **Human-in-the-loop control:** ensure the system only detects and informs; every consequence remains a decision made by the teacher.

Scope of the Project

The system is designed to function as an assistive decision-support layer for instructors, not as a replacement for them. Its scope includes:

- **Browser-side signal collection:** tab visibility, window focus and copy/paste/cut events from the browser, and gaze, head pose and face presence from the webcam through MediaPipe.
- **Personal baseline and deviation scoring:** a calibration phase followed by scoring relative to that baseline.
- **Evidence-to-risk pipeline:** episodes, fusion, features and a risk band with reasons.
- **Session logging:** storage of signals, evidence, features and risk results in SQLite for later review and for building the training dataset.

System Overview

The system has three parts:

1. **Client side (student's browser):** a Browser Signal Collector and a Vision/Gaze Collector run in the page. After a guided calibration, only small derived numbers leave the browser.
2. **Server side:** an Express backend receives the derived values and stores them in SQLite. The evidence, fusion, feature and risk stages convert them into an explainable risk band.
3. **Instructor side: the implemented Teacher Dashboard lists Medium and High sessions with their contributing signals, episode counts and review actions for human review.**

Advantages of the System

- **Fairer baseline:** each student is compared with their own behavior, so natural variation becomes part of their personal norm.
- **Privacy by design:** raw video is processed in the browser and discarded after each frame; only derived values are transmitted.
- **Explainability:** the scoring model is simple enough that every contribution can be shown to the instructor.
- **Reduced false alerts from isolated events:** evidence is grouped into episodes before it affects the score.
- **Low deployment cost:** it needs only a webcam and a browser, with no extra hardware.

Applications of the Project

- Online semester and internal examinations in colleges and universities.
- Remote certification and entrance tests where an instructor reviews flagged sessions afterwards.
- Research on fair and privacy-preserving proctoring, using the logged evidence as a dataset.

Expected Outcomes

- A working browser-based pipeline that collects signals, calibrates each student and produces an explainable risk band.
- A labeled, scenario-based dataset recorded with the application itself.
- An interpretable Logistic Regression classifier trained on that dataset and compared with a fixed-threshold baseline.
- A measured, offline comparison of detector confidence across demographic groups on FairFace.

Future Enhancements

- A Teacher Dashboard with per-signal explanation chips and a review workflow.
- Fullscreen-exit detection and optional copy/paste blocking.
- A larger and more diverse participant pool and a reviewer trial to measure automation bias.

1.2 SCOPE AND OBJECTIVE

Purpose of the Study

The purpose of this study is to design, implement and evaluate a proctoring layer that scores each student against their own recorded baseline, keeps raw camera data on the student's device and gives the instructor an explanation instead of an unexplained alert. The study also aims to measure, rather than assume, how much the underlying face detector's confidence varies across demographic groups.

General Objective

The general objective is to build a privacy-preserving, explainable and individually calibrated risk-scoring system for online exams in which a human instructor makes every final decision.

Specific Objectives

- To implement in-browser extraction of gaze, head pose and face presence using MediaPipe Face Landmarker.
- To implement a guided calibration that records a personal baseline for each student.
- To group signals into temporal episodes using a 1.5-second gap rule and fuse them into model-ready features.
- To train an interpretable Logistic Regression classifier on a self-recorded, scenario-labeled dataset.
- To convert the classifier output into Low / Medium / High bands with accommodation-shifted thresholds.
- To audit MediaPipe and RetinaFace on FairFace offline and report the measured confidence disparity.

CHAPTER 2. LITERATURE REVIEW

2.1 LITERATURE OVERVIEW

Automated proctoring research can be grouped into three lines of work. The first line builds multimodal monitoring systems that combine a webcam with other sensors to check identity, gaze, speech and surrounding objects. The second line tries to reduce the privacy cost of video-based proctoring, for example by hiding the face while keeping the eyes visible for gaze tracking. The third line studies the fairness and ethics of automated exam supervision and of face-analysis technology in general.

Atoum et al. describe a multimodal system that combines a webcam with a wearable camera. Yaqub, Mohanty and Suleiman address privacy by blurring or masking the face and detecting when the gaze leaves the screen, using a per-examinee calibration model. AutoOEP, a recent preprint, uses a dual-camera setup with identity verification, head pose, gaze, mouth movement, an object detector and an LSTM that outputs a cheating probability. A patent assigned to ProctorU describes automated proctoring with identity validation, room scanning, recording and behavior analysis.

On fairness, Buolamwini and Gebru demonstrated accuracy disparities across demographic groups in commercial face analysis, which motivates auditing any face detector that is used in a high-stakes setting. Coghlan, Miller and Paterson discuss the ethical concerns of online exam supervision, including privacy intrusion. Commercial platforms such as Proctorio, Examity and Proctor360 provide risk indicators or advertise behavioral baselines, but the technical detail needed for independent evaluation is not publicly available.

From this review the gap addressed by GlassBox is clear: the surveyed systems either depend on continuous video, or use calibration only to improve tracking accuracy, or do not explain their output. GlassBox combines a personal behavioral baseline, local processing and an interpretable score in one browser-based system, and it evaluates the fairness of its detector explicitly.

2.2 LITERATURE SURVEY

Table 2.1: Summary of the literature survey

S.No	Author and Title	Methodology	Limitations	GlassBox contribution
1	Y. Atoum et al., "Automated online exam proctoring," IEEE Trans. Multimedia, 2017	Webcam plus wearable camera; analyzes identity, gaze, speech, surrounding text, window activity and phone presence.	Needs extra wearable hardware and continuous capture; privacy-intrusive.	Uses only the built-in webcam and browser; raw video stays on the device.

S.No	Author and Title	Methodology	Limitations	GlassBox contribution
2	W. Yaqub, M. Mohanty, B. Suleiman, "Proctoring online exam using eye tracking," SECRYPT, 2023	Face hidden by blurring or masking while eyes stay visible; gaze tracking with a per-examinee calibration model detects gaze leaving the screen.	Calibration is used to improve gaze-to-screen mapping; small in-house video dataset; video is still reviewed.	Reuses the idea of per-student calibration as a behavioral baseline for all signals, adds explainable scoring.
3	AutoOEP – A multi-modal framework for online exam proctoring, arXiv, 2025	Dual cameras, ArcFace identity check, head pose, gaze, mouth movement, fine-tuned YOLO detector and an LSTM; reports 90.7% accuracy on a custom dataset.	Continuous dual-camera video; an LSTM output is hard to explain; not privacy-preserving.	Interpretable Logistic Regression and local processing instead of continuous video.
4	M. Jaeh, J. Morgan, A. Millin, "System for online automated exam proctoring," US Patent 12,039,887 B2 (ProctorU)	Identity validation, room scanning, audio/video recording and behavior analysis, including thresholds calibrated to a baseline of the test-taker's environment.	A patent description, not a peer-reviewed evaluation; relies on recording and room scans.	Baseline is the student's own behavior, not the environment; no recording is transmitted.
5	J. Buolamwini, T. Gebru, "Gender Shades," FAT*, 2018	Audited commercial face analysis and showed accuracy disparities across intersectional demographic groups.	Addresses gender classification, not proctoring.	Motivates auditing the face detector used here across groups.
6	S. Coghlan, T. Miller, J. Paterson, "Good proctor or Big Brother?," Philosophy & Technology, 2021	Ethical analysis of online exam supervision technologies, including privacy intrusion.	Conceptual analysis; no technical system.	Privacy-by-design and human-in-the-loop decisions.
7	K. Kärkkäinen, J. Joo, "FairFace," WACV, 2021	Face dataset balanced across race, gender and age groups for bias measurement.	Annotates race, not skin tone; a benchmark, not a proctoring system.	Used for the offline detector-confidence audit, reported per race group as a proxy.
8	J. Deng et al., "RetinaFace," CVPR, 2020	Single-stage multi-level face localization with joint box, landmark and dense-regression heads.	Heavier model; needs server support for real-time use.	Used only offline, unmodified, as an independently trained comparison detector.
9	V. Bazarevsky et al., "BlazeFace," 2019; I. Grishchenko et al., "Attention Mesh," 2020	Lightweight face detector and attention-based landmark network that underlie MediaPipe Face Landmarker.	Pretrained; the training-data composition is not under our control.	Used unmodified in the browser; its behavior is audited rather than assumed.

CHAPTER 3. SYSTEM DESIGN

3.1 EXISTING SYSTEM

Existing proctoring systems fall into three types: live human proctoring over video, record-and-review systems that store the whole session for later inspection, and automated systems that analyze the video stream with computer vision. Automated systems usually apply one set of fixed thresholds to every examinee, and the final score is often presented without an explanation.

The main limitations of existing systems are:

- A single fixed standard treats natural individual differences as suspicious behavior.
- Face-analysis components can show demographic performance gaps, which may create unequal false alerts.
- Continuous video and audio recording creates privacy and data-protection obligations.
- Scores are opaque, so students and instructors cannot see which behavior produced an alert.
- Some systems apply consequences automatically without human review.

3.2 PROPOSED SYSTEM

GlassBox is proposed as a layer that sits alongside an exam platform. It collects browser and webcam signals on the client, calibrates each student against their own baseline, groups evidence into episodes, fuses it into features and converts it into a risk band with reasons. A human instructor reviews flagged sessions and makes every decision. Table 3.1 compares the proposed design with the existing approach.

Table 3.1: Existing approach versus GlassBox (design)

Criterion	Typical existing system	GlassBox
Personal calibration baseline	No (fixed population thresholds)	Yes – guided baseline per student
Raw video stays on the device	Usually no (continuous capture)	Yes – processed in the browser and discarded
Signal-level explanation	Limited or not published	Yes – per-signal contribution
Measured detector-fairness audit	Not reported	Completed – FairFace, offline
Human review of every consequence	Varies; some act automatically	Yes – no automatic penalty
Accommodation handling	Often a separate rule or none	Shifted thresholds in the same formula

3.3 SYSTEM ARCHITECTURE

Figure 1 shows the architecture. The client stage runs in the student's browser and never transmits raw biometric data. A dashed privacy boundary separates it from the server stage, which converts the small derived values into an explainable risk band and stores them. Colors are used consistently across the figures: orange for client-side, blue for pipeline processing, purple for the machine-learning model, green for storage and teal for the human reviewer.

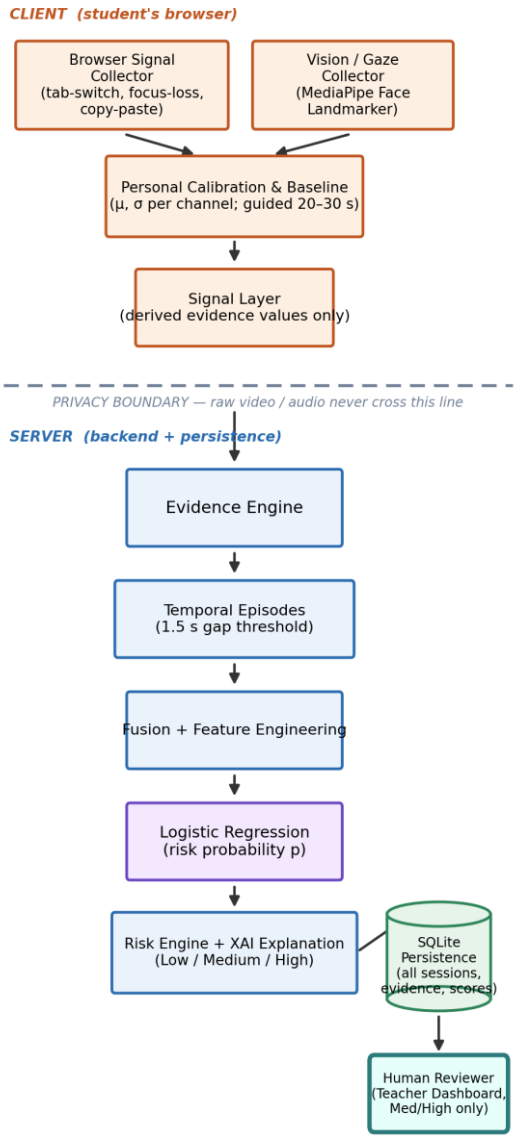


Figure 1: GlassBox end-to-end system architecture

3.3.1 Process Flow

The end-to-end flow of one examination session is as follows:

1. The student opens the exam link and grants webcam permission.
2. The Vision/Gaze Collector starts MediaPipe Face Landmarker in the browser and the Browser Signal Collector starts listening for tab, focus and clipboard events.
3. The student completes the guided calibration; the system computes the personal baseline (mean and standard deviation per channel).
4. During the exam each frame is processed locally and discarded; only derived values (gaze deviation, head pose, face presence) and browser events are emitted as signals.
5. Signals are sent to the backend, where the Evidence Engine interprets them and merges evidence that is at most 1.5 seconds apart into episodes.
6. The Fusion and Feature stage combines concurrent episodes of different types and builds the feature vector.
7. The Risk Engine converts the features and evidence into a Low / Medium / High band and a list of reasons.
8. Signals, evidence, features, fusion events and risk results are stored in SQLite.
9. Medium and High sessions are made available to the instructor, who reviews the evidence and decides; no exam is terminated automatically.

3.3.2 Module Description

The system is divided into ten modules, listed in Table 3.2 together with their layer, function and current status. The status column reflects the code in the project repository at the time of writing.

Table 3.2: Modules of GlassBox

No	Module	Layer	Function	Input	Output	Status
1	Browser Signal Collector	Client	Captures tab hidden/visible, window focus lost/gained (with duration) and copy/paste/cut events.	Browser events	Signal objects	Implemented
2	Vision / Gaze Collector	Client	Runs MediaPipe Face Landmarker per frame; derives gaze, head pose (yaw, pitch, roll) and face presence/absence.	Webcam frames	Numeric signals	Implemented

No	Module	Layer	Function	Input	Output	Status
3	Personal Calibration and Baseline	Client	Guided sequence that records the student's own gaze and head-pose statistics.	Gaze, head pose	Baseline μ, σ	Implemented
4	Evidence Engine	Pipeline	Interprets raw signals and builds evidence records and episodes.	Signals	Evidence	Implemented
5	Temporal Episode Grouping	Pipeline	Merges evidence separated by ≤ 1.5 s into one episode.	Evidence	Episodes	Implemented
6	Fusion and Feature Engineering	Pipeline	Fuses concurrent episodes and builds the feature vector.	Episodes	Features	Implemented
7	Logistic Regression Classifier	ML	Interpretable model that converts features to a risk probability.	Features	Probability p	Implemented and trained
8	Risk Engine and Explanation	Pipeline	Converts evidence to a Low/Medium/High band with reasons. Currently an equal-weight rule baseline.	Evidence, fusion	Band + reasons	Implemented (rule baseline)
9	SQLite Persistence	Server	Express REST API storing sessions, signals, evidence, features, fusion events and risk results.	JSON records	Database rows	Implemented
10	Teacher Dashboard	Instructor	Lists Medium/High sessions with reasons for human review.	Stored risk results	Review view	Implemented

3.3.3 Models Used

GlassBox does not train a new vision model. It uses two pretrained, unmodified vision models and one small classifier that the team trains itself (Table 3.3 and Figure 2). MediaPipe Face Landmarker is the only model that runs in the live pipeline. RetinaFace is used only in the separate offline fairness audit and is never run on real student sessions.

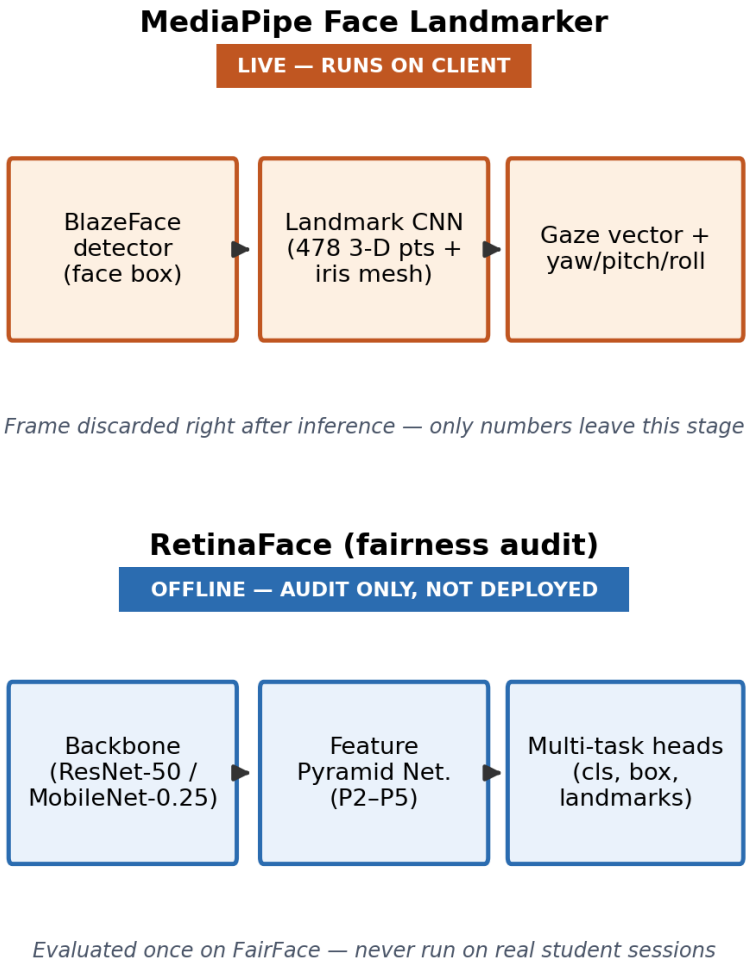


Figure 2: Internal structure of the pretrained vision models used

Table 3.3: Models used in the GlassBox pipeline

Model	Role	Training status	Structure
MediaPipe Face Landmarker	Live gaze and head-pose extraction in the browser	Pretrained, used unmodified	Face detector + landmark network
RetinaFace	Offline fairness audit only	Pretrained, used unmodified	Backbone + FPN + multi-task heads
Logistic Regression	Risk scoring (Low/Medium/High)	To be trained by the team	Linear layer + sigmoid

MediaPipe Face Landmarker chains a lightweight face detector with a landmark network that predicts 478 three-dimensional landmarks, including the iris. Gaze direction is derived from the iris landmarks relative to the eye contour, and head pose (yaw, pitch and roll) is derived from the 4×4 facial transformation matrix that the library returns. RetinaFace uses a ResNet-50 or MobileNet-0.25 backbone, a Feature Pyramid Network and multi-task heads for face score,

bounding box, landmarks and dense regression. Because it is trained independently of MediaPipe, agreement between the two on the same fairness pattern is stronger evidence than either model alone.

3.4 HARDWARE REQUIREMENTS

Table 3.4 lists the recommended hardware. These are design recommendations; the minimum configuration will be confirmed after the in-browser frame rate has been measured on target devices.

Table 3.4: Hardware requirements

Component	Requirement
Student device	Laptop or desktop with a built-in or external webcam (720p or higher recommended)
Processor	Modern multi-core CPU capable of running MediaPipe in a browser
Memory	8 GB RAM recommended
Display and input	Standard display, keyboard and mouse
Network	Stable broadband connection to reach the backend
Server / development machine	Any machine able to run Node.js and SQLite

3.5 SOFTWARE REQUIREMENTS

Table 3.5: Software requirements

Component	Technology
Front end	React 19 with Vite
Vision library	MediaPipe Tasks Vision (Face Landmarker), @mediapipe/tasks-vision
Back end	Node.js with Express 5 and CORS middleware
Database	SQLite through the better-sqlite3 library
Model training	Python with scikit-learn (LogisticRegression)
Fairness audit	RetinaFace implementation and the FairFace dataset
Browser	Chromium-based browser with webcam permission
Tools	Visual Studio Code, Git and GitHub

CHAPTER 4. IMPLEMENTATION

4.1 METHODOLOGY

Overview

The methodology converts raw browser and camera observations into a risk band in five stages: personal calibration, deviation scoring, temporal episode formation, fusion and feature construction, and risk scoring with explanation. The formulas of each stage are given below so that the path from raw signal to risk band can be audited.

Personal Calibration and Deviation Score

During calibration, N frames are collected for each channel j (gaze coordinates and head-pose angles). The baseline mean and standard deviation of channel j are

$$\mu_j = (1/N)\,\Sigma\, x_{ij}\,, \quad \sigma_j = \sqrt{[\,(1/N)\,\Sigma\,(x_{ij} - \mu_j)^2\,]} \qquad (1)$$

Every later observation x_j(t) is converted into a personal deviation score instead of a raw value:

$$d_j(t) = |\,x_j(t) - \mu_j\,| \,/\, (\,\sigma_j + \varepsilon\,) \qquad (2)$$

where $\varepsilon > 0$ is a small constant that prevents division by zero when a channel has almost no variance during calibration. Because d_j(t) is expressed in units of the student's own variability, a naturally fidgety student and a naturally still student are held to the same relative standard rather than the same absolute threshold. This is the core fairness mechanism of GlassBox.

Temporal Episodes

Raw signals are not treated as evidence of misconduct on their own. Two consecutive evidence events at times t_i and t_(i+1) belong to the same episode if

$$t_{(i+1)} - t_i \le \Delta t\,, \quad \Delta t = 1.5\ s \qquad (3)$$

and a new episode begins otherwise. Each episode is summarized by its duration, its strength and the type of signal it contains. In the code this threshold is the parameter episodeGapMs = 1500.

Fusion and Features

The fusion stage looks at episodes of different types that overlap in time and produces a fusion score with a confidence value and a count of concurrent signal types. The feature stage then summarizes, for each episode type (gaze, head pose, focus loss and face absence), the frequency, total duration and strength of the episodes, and adds the fusion score, fusion confidence, the concurrence count and a combined behavioral-activity value.

Risk Scoring – Current Rule-Based Baseline

The prototype currently uses a transparent, equal-weight rule baseline. For each of five components k (focus loss, face absence, gaze deviation, head-pose deviation and multi-signal fusion) a component score s_k in [0, 1] is computed, where for an episode type s_k is the sum of its episode strengths divided by 3 and capped at 1. The risk score is

$$S = 100 \times clip(\Sigma_k w_k s_k , 0 , 1) , \quad w_k = 0.20 \text{ for all } k \qquad (4)$$

$$level(S) = LOW \text{ if } S < 30;\ MEDIUM \text{ if } 30 \leq S < 60;\ HIGH \text{ if } S \geq 60 \qquad (5)$$

This baseline is deliberately simple. The weights are fixed and are not learned from data; replacing them by learned weights is the purpose of the Logistic Regression stage.

Risk Scoring – Trained Logistic Regression Classifier

The only model that the team trains itself is a Logistic Regression classifier. It is a single linear layer of size k followed by a sigmoid, with no hidden layers, so each weight is directly interpretable as the influence of one behavioral feature:

$$z = w^T x + b = \Sigma w_i x_i + b \qquad (6)$$

$$p = \sigma(z) = 1 / (1 + e^{(-z)}) \qquad (7)$$

where p in (0, 1) is the estimated risk probability. Training minimizes the binary cross-entropy over M labeled examples with an L2 penalty on the weights:

$$L(w, b) = -(1/M) \Sigma [y_j \log p_j + (1 - y_j) \log(1 - p_j)] + (\lambda/2) \|w\|_2^2 \qquad (8)$$

where $y_j \in \{0,1\}$ is the binary training label and $\lambda \geq 0$ is the regularization strength, selected using training-only cross-validation. The L2 term penalizes large weights and may improve generalization; its benefit must be verified empirically.

Risk Bands and Accommodation Shift

The trained classifier converts estimated probability p into a risk band using validated operating thresholds τ_1 and τ_2. The transparent rule-based baseline separately uses a 0–100 score with cut-offs of 30 and 60 in Equation (5); these two scoring systems are evaluated separately and are not interchangeable.

$$band(p) = Low \text{ if } p < \tau_1;\ Medium \text{ if } \tau_1 \leq p < \tau_2;\ High \text{ if } p \geq \tau_2 \qquad (9)$$

Accommodated students receive individually shifted thresholds using the same rule rather than a separate one:

$$\tau_1' = \tau_1 + \Delta, \quad \tau_2' = \tau_2 + \Delta \qquad (10)$$

where Δ is a small, disclosed, instructor-approved offset. The scoring logic stays identical for every student; only the operating point moves.

Explanation

Because the model is linear, the contribution of feature i is $c_i = w_i x_i$ and its relative share can be shown to the reviewer:

$$c_i = w_i x_i, \quad s_i = |c_i| / \Sigma_j |c_j| \qquad (11)$$

The features with the largest normalized contributions can be shown as reasons for a Logistic Regression prediction. If $\Sigma_j |c_j| = 0$, the relative shares are undefined and should be reported as unavailable rather than divided by zero. In the current rule-based baseline, explanations are generated from component scores, for example: “Gaze deviation was sustained at a notable level.”

Evaluation Metrics

With true and false positives and negatives (TP, FP, TN, FN), the classification metrics are

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN) \qquad (12)$$

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall}) \qquad (13)$$

ROC-AUC is the area under the true-positive-rate versus false-positive-rate curve over all thresholds. For the fairness audit, if m_g is the mean detector confidence in group g, the disparity is

$$\Delta_{conf} = \max_g m_g - \min_g m_g \qquad (14)$$

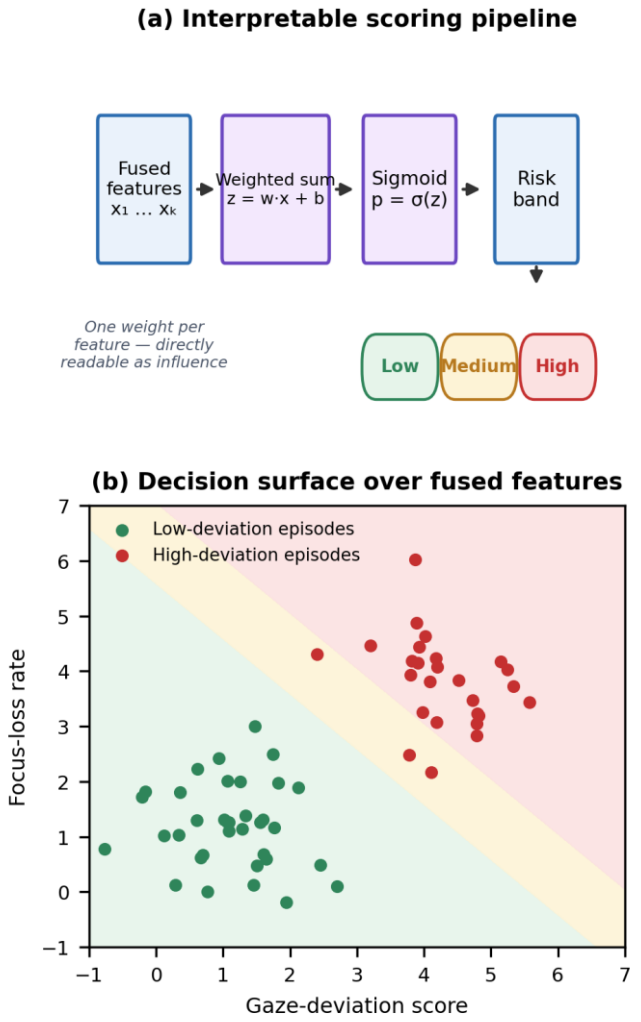


Figure 3: Interpretable risk-scoring pipeline and decision surface

4.2 IMPLEMENTATION

GlassBox is implemented as a React and Vite web application with an Express backend and an SQLite database. The source code is organized by module (src/modules, src/evidence, src/risk, server). Table 4.1 summarizes the implemented system components.

Table 4.1: Implementation status of the GlassBox components

Component	Status
Browser signal collection (tab, focus, copy/paste/cut)	Implemented
Fullscreen-exit detection	Outside implemented scope
MediaPipe in-browser gaze, head pose, face presence	Implemented
Guided personal calibration (client side)	Implemented
Evidence engine and temporal episodes (1.5 s)	Implemented
Fusion and feature engines	Implemented
Risk engine with explanation (equal-weight rule baseline)	Implemented

Component	Status
Express backend and SQLite persistence	Implemented
Logistic Regression classifier	Implemented and trained
Teacher Dashboard	Implemented
Offline FairFace fairness audit	Completed
Reviewer trial	Reported

4.2.1 Browser Signal Collection Module

The module (browserSignalCollector.js with the React hook useBrowserSignals.js) registers listeners on the page and converts each event into a signal object with a type, source, value and confidence. The signals produced are TAB_HIDDEN and TAB_VISIBLE from the visibility-change event, FOCUS_LOST and FOCUS_GAINED from the window blur and focus events (the focus-gained signal carries the duration of the preceding focus loss), and COPY, PASTE and CUT from the clipboard events.

Clipboard events are recorded as contextual signals; they are not blocked and are not directly treated as evidence of misconduct. The browser monitoring module also records the supported visibility and focus events. Fullscreen-exit detection is outside the implemented scope.

4.2.2 Vision, Gaze and Calibration Module

The module (VisionModule.jsx, visionTracker.js, gazeMath.js and headPoseMath.js) loads MediaPipe Face Landmarker through @mediapipe/tasks-vision and processes the webcam stream inside the browser. For each frame it produces the face landmarks and the facial transformation matrix. Gaze is computed from the iris landmarks relative to the eye contour, and head pose is computed from the transformation matrix as yaw, pitch and roll in degrees. Face presence, absence and a face-quality value are emitted as signals. The video frame is discarded after inference and only the derived numbers are used.

Calibration is implemented on the client in visionTracker.js and has three phases: a natural-behavior phase of 8 seconds, a guided phase of 18 seconds in which the student looks at nine targets (center, top-left, top, top-right, right, bottom-right, bottom, bottom-left and left), and a final natural-behavior phase of 7 seconds, a total of 33 seconds. The collected gaze and head-pose samples and a quality measure are used to form the baseline of Equation 1.

4.2.3 Evidence Engine and Temporal Episodes

The EvidenceEngine class receives signals and builds evidence records of four episode types: FOCUS_LOSS_EPISODE, FACE_ABSENCE_EPISODE, GAZE_DEVIATION_EPISODE and HEAD_POSE_DEVIATION_EPISODE. It keeps a sliding window (windowMs = 10000), a decay period (decayMs = 30000) and the episode gap (episodeGapMs = 1500) of Equation 3. An active gaze or head-pose episode is extended while new deviations arrive within the gap and is closed, with its duration and strength, when the gap is exceeded. A single frame therefore cannot raise the risk directly.

4.2.4 Fusion and Feature Engineering

The fusion engine (fusionEngine.js) combines concurrent episodes of different types into a fusion score and a confidence value. The feature engine (featureEngine.js) then produces, for gaze, head pose, focus loss and face absence, the frequency, total duration and strength of episodes, together with the fusion score, the fusion confidence, a concurrence count and a combined behavioral-activity value. These outputs form the feature vector that will be used to train the classifier.

4.2.5 Risk Engine and Explanation

The RiskEngine class (riskEngine.js) computes the risk score of Equation 4 from the evidence history and the latest fusion result, maps it to LOW, MEDIUM or HIGH using Equation 5 and builds a list of reasons. It is intentionally not driven by raw signals: risk is calculated only from completed evidence and fusion results. The code comments state explicitly that this equal-weight version is a transparent baseline and that Logistic Regression will later learn the weights from data.

4.2.6 Backend and Persistence

The backend (server/server.js and server/database.js) is an Express 5 application that uses better-sqlite3 to store data in a local SQLite file. The front end sends records through a persistence manager and a small API client. The main REST endpoints are listed in Table 4.2.

Table 4.2: Main REST endpoints of the backend

Endpoint	Method	Purpose
/api/health	GET	Check that the backend is running
/api/exams	POST	Create or register an exam
/api/students	POST	Register a student for a session
/api/sessions	POST	Start an exam session (further endpoints manage and retrieve sessions)

Endpoint	Method	Purpose
/api/signals	POST	Store raw signals
/api/evidence	POST	Store evidence episodes
/api/features	POST	Store the computed feature vector
/api/fusion	POST	Store fusion events
/api/risk	POST	Store the risk result (score, level, reasons)

The database contains the tables exams, students, exam_sessions, signals, evidence, features, fusion_events and risk_results. The local database file contains only development and test data and must not be committed to the public repository; it is excluded through the .gitignore of the data-collection branch.

4.2.7 Teacher Dashboard (Design)

The implemented Teacher Dashboard presents stored session results in a review-oriented interface. Each row shows the student identifier, risk band, top contributing signal, episode count and a Review action. The dashboard prioritizes Medium and High sessions for human review and displays derived evidence and explanations rather than raw video. The figure should be replaced with a screenshot captured from the running application for final submission.

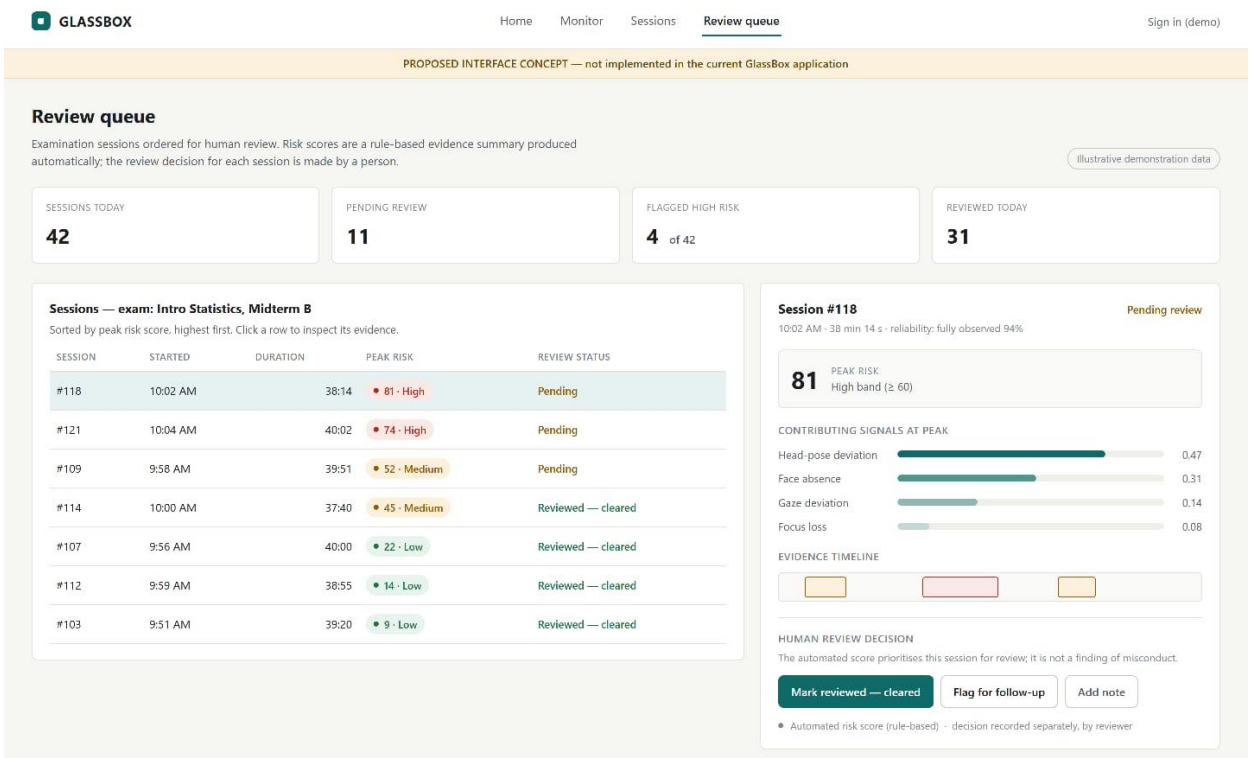


Figure 4: Teacher Dashboard interface

4.2.8 Logistic Regression Training and Evaluation Pipeline

The completed training pipeline exports feature vectors and session labels from SQLite to CSV, merges the labeled sessions recorded by the team and trains a scikit-learn LogisticRegression model with an L2 penalty. Evaluation keeps sessions grouped so that episodes from one session do not leak across training and test folds; a leave-one-person-out variant is used to examine generalization across participants. The evaluation compares a fixed-threshold baseline on raw signals with the calibrated-feature Logistic Regression model, with a raw-feature classifier included where available to isolate the contribution of personal calibration. The trained model provides learned feature weights, while the equal-weight rule remains a transparent baseline.

4.3 DATASET

Video-only proctoring datasets cannot represent browser signals such as tab switches and focus loss, and they would not support the explainability goal. GlassBox therefore uses data generated with the application itself. Three team members each record scripted sessions in which they perform a chosen behavior. The data-collection interface records, for every session, a condition label such as NORMAL, LOOK_AWAY, HEAD_TURN, FACE_ABSENT, FOCUS_LOSS, MULTI_SIGNAL or OTHER. Sessions are labeled at the session level, which avoids frame-by-frame annotation, and each member's local SQLite database is exported to CSV and merged before training.

Development and test sessions created during implementation are excluded from the labeled training/evaluation dataset. The dataset represents scripted behavior from a small participant pool; therefore, its results should be interpreted as a controlled prototype evaluation rather than evidence of generalization to all examination settings. Report the final session and episode counts from the exported dataset in the dataset summary table before submission.

The completed fairness audit uses the FairFace dataset offline. MediaPipe and RetinaFace are evaluated on the same images without fine-tuning, comparing detection rate and mean confidence among detected faces across annotated groups. Confidence is interpreted alongside detection rate because confidence among detected faces alone can conceal missed detections. FairFace annotates race, gender and age rather than skin tone, so race-group comparisons are treated as a limited proxy analysis and not a direct measurement of skin tone.

4.4 PROJECT TIMELINE

Figure 5 summarizes project activities from July to October 2026. The timeline records completed implementation, model training, fairness auditing and review activities; the chart should be updated to match the final project dates before submission.

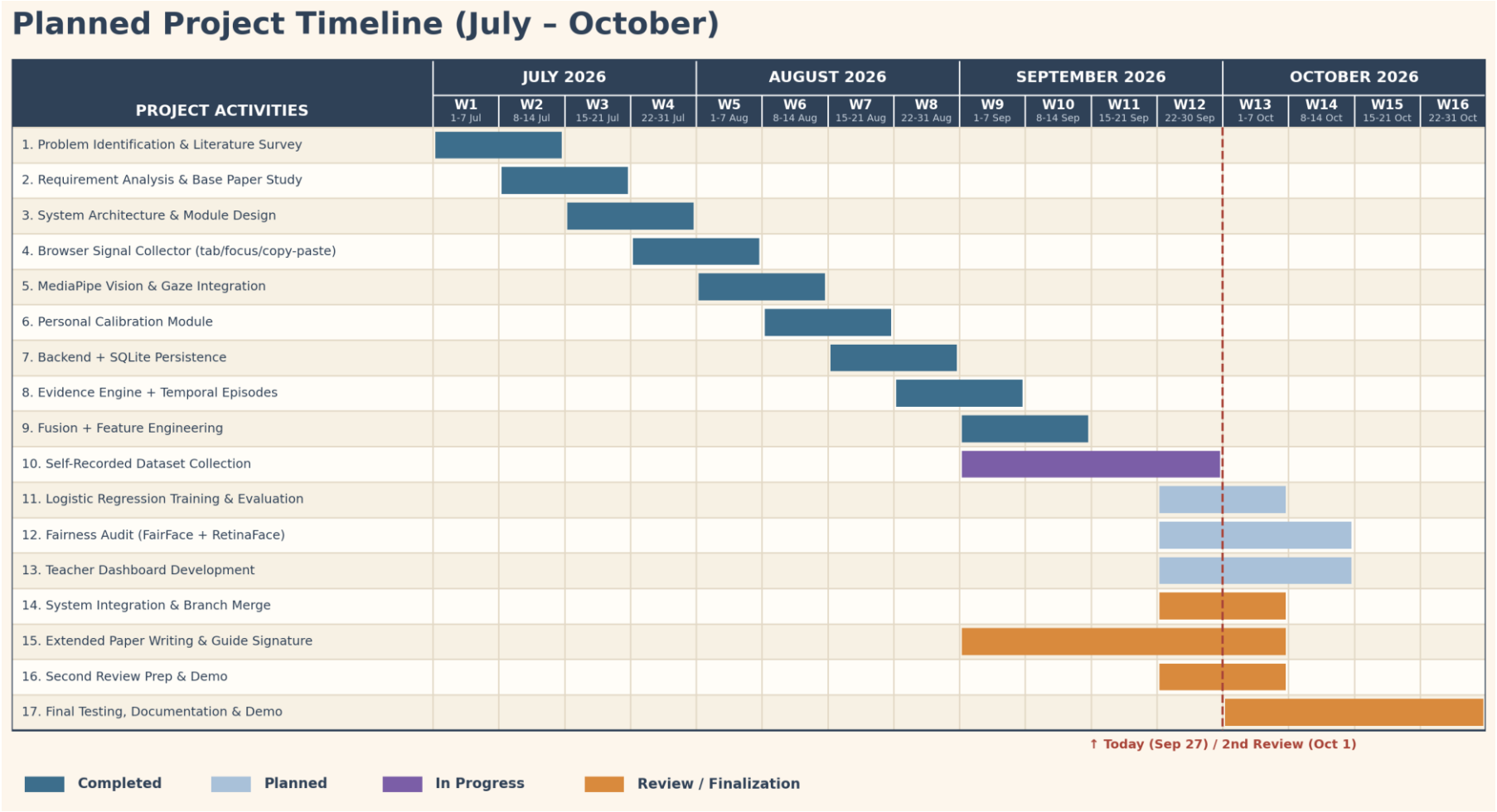


Figure 5: Project timeline (Gantt chart), July–October 2026

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

GlassBox reframes exam-proctoring fairness as a calibration problem. Instead of adding a new detector or a separate rule for accommodated students, it scores every student against their own recorded baseline, keeps the scoring model interpretable and keeps raw camera data inside the browser. The formulas of Chapter 4 make the path from a raw signal to a risk band auditable, and the design never terminates an exam automatically.

The completed system includes browser signal collection, in-browser MediaPipe extraction, three-phase personal calibration, evidence and temporal episode formation with a 1.5-second gap rule, fusion and feature construction, an explainable equal-weight baseline, a trained Logistic Regression classifier, the Teacher Dashboard, and an Express/SQLite persistence layer. The FairFace detector audit has also been carried out offline. Quantitative claims in this report are restricted to the actual values entered in the results tables; no metric should be inferred from the implementation description alone.

5.2 FUTURE WORK

- Expand the scripted-session dataset with more participants and varied recording conditions, and report the resulting session and episode counts.
- Evaluate additional feature sets and grouped validation strategies to test the stability of the trained Logistic Regression classifier against the equal-weight baseline.
- Fix the final risk thresholds on validation data and implement the accommodation shift in the Risk Engine.
- The implemented Teacher Dashboard supports review of Medium- and High-risk sessions and displays their explanations.
- Add fullscreen-exit detection and decide whether clipboard events should be blocked or only logged.

- Document the completed FairFace audit of MediaPipe and RetinaFace, including measured confidence disparity by annotated race group.
- Measure the in-browser frame rate, per-frame latency and data sent to the server per minute on target devices.
- Run a reviewer trial to measure review time and whether reviewers catch wrong or borderline flags, as a check for automation bias.
- Extend the participant pool to improve diversity and reduce the variance of cross-validation estimates.

5.3 RESULTS

Prototype screens

Figures 6–8 present the application views used to document the implemented prototype. Insert genuine screenshots captured from the running application in these locations. The screenshots should show the calibration workflow, live monitoring view, and stored session records.

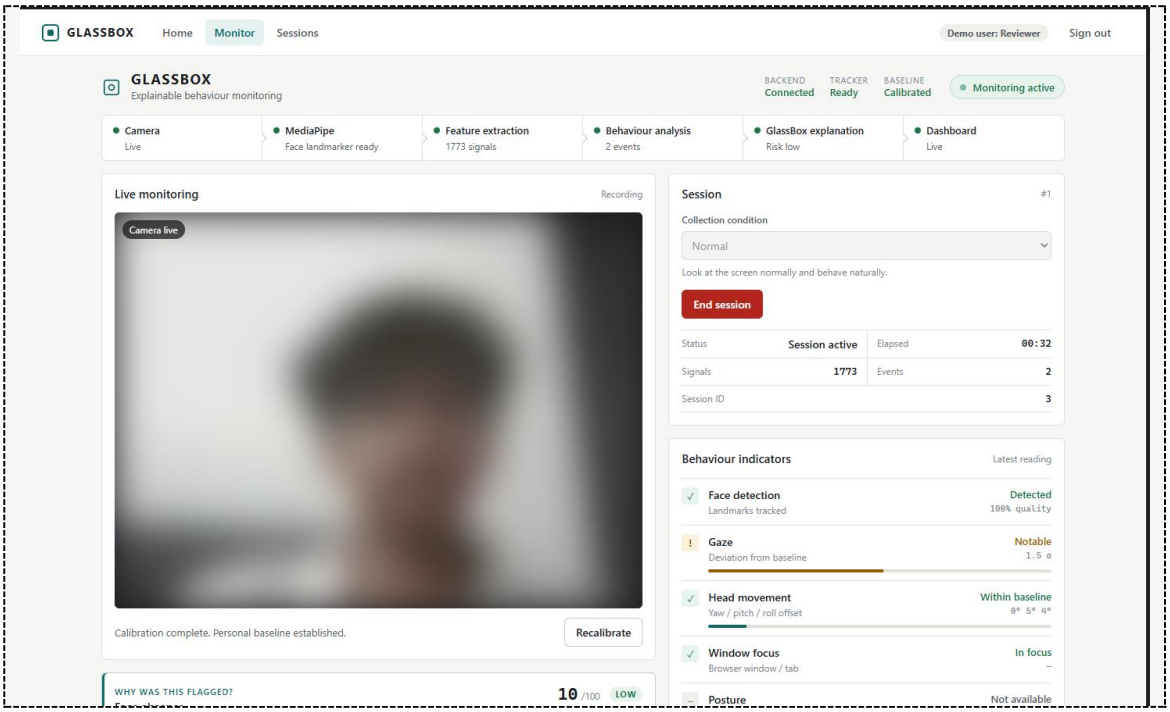

GLASSBOX
Home
Monitor
Sessions
Demo user: Reviewer
Sign out
GLASSBOX
Explainable behaviour monitoring
BACKEND
Connected
TRACKER
Ready
BASELINE
Calibrated
Monitoring active
Camera
Live
MediaPipe
Face landmarker ready
Feature extraction
1773 signals
Behaviour analysis
2 events
GlassBox explanation
Risk low
Dashboard
Live
Live monitoring
Recording
Camera live
Calibration complete. Personal baseline established.
Recalibrate
WHY WAS THIS FLAGGED?
10 /100
LOW
Session
#1
Collection condition
Normal
Look at the screen normally and behave naturally.
End session
Status
Session active
Elapsed
00:32
Signals
1773
Events
2
Session ID
3
Behaviour indicators
Latest reading
Face detection
Landmarks tracked
Detected
100% quality
Gaze
Deviation from baseline
Notable
1.5 σ
Head movement
Yaw / pitch / roll offset
Within baseline
0° 5° 4°
Window focus
Browser window / tab
In focus
–
Posture
Not available

Figure 6: Calibration screen of the prototype

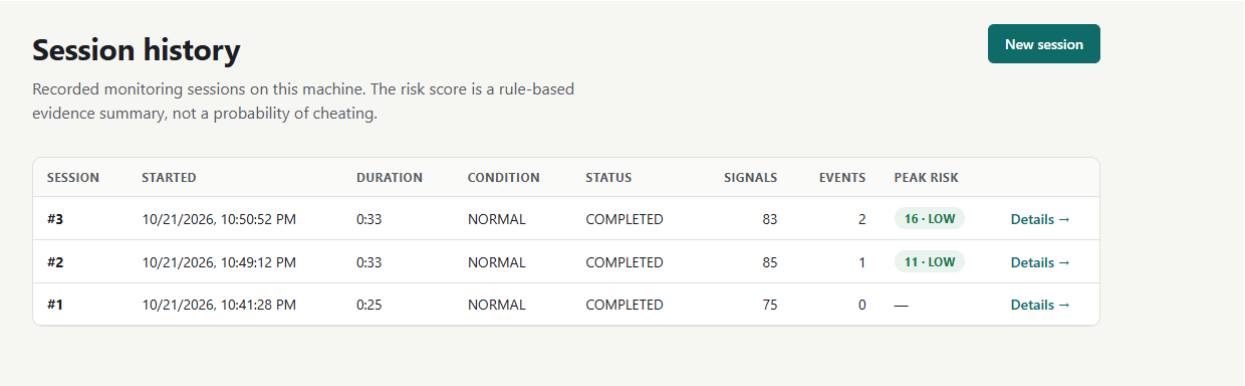


Figure 7: Live monitoring view showing signals, evidence and risk

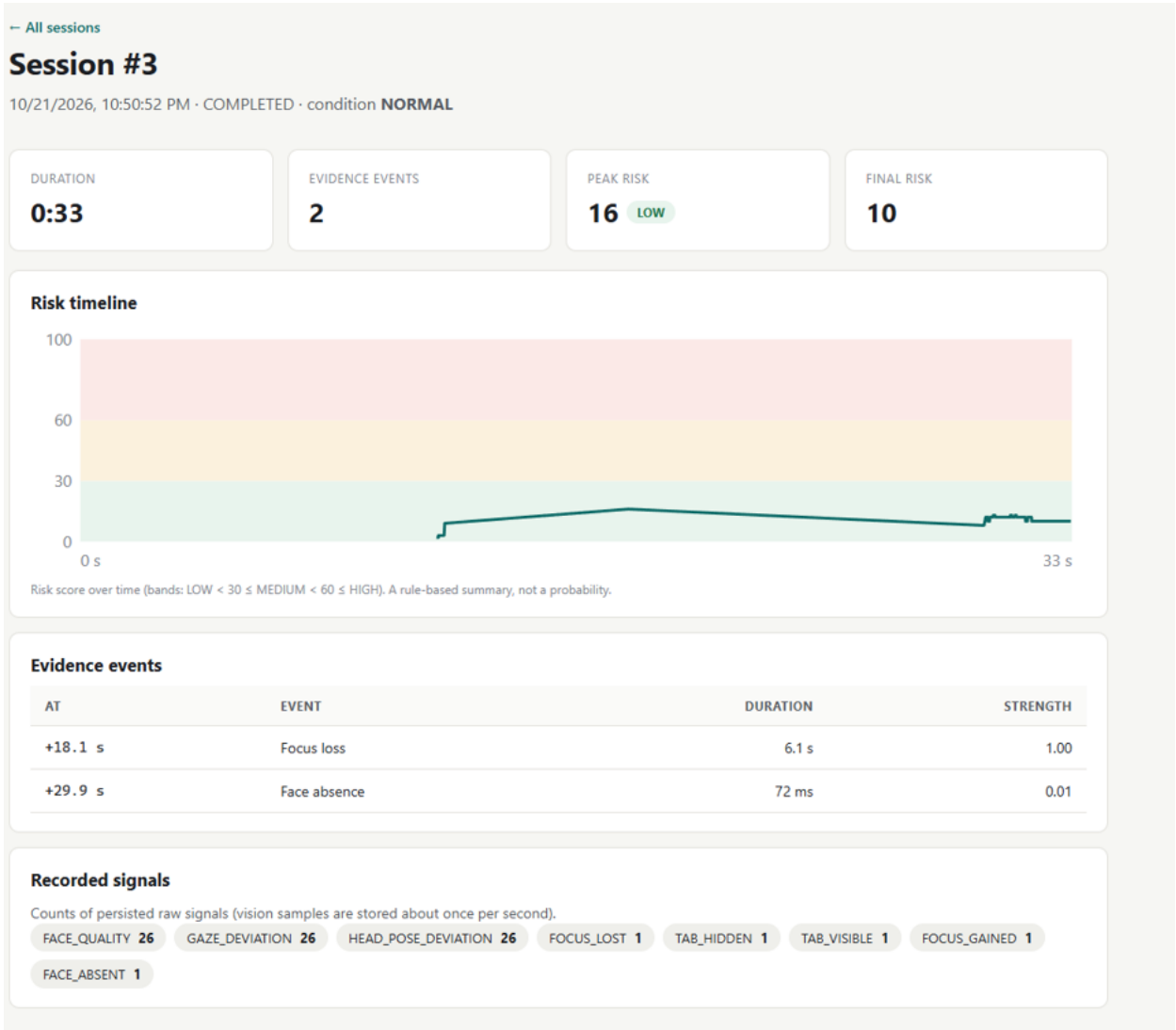


Figure 8: Live monitoring view showing signals, evidence and risk

Evaluation and validation results

The completed evaluation covers two parts: classification performance and detector fairness. Classification is assessed using session-grouped validation to avoid placing sessions from the same participant in both training and validation folds. The reported classification measures are

precision, recall, F1-score, ROC-AUC, false-positive rate and false-negative rate. The FairFace audit compares sample counts, face-detection rates and mean detector confidence across annotated race groups for MediaPipe and RetinaFace. FairFace labels race rather than skin tone, so these group comparisons are treated as a limited proxy and not as a complete measure of fairness. The numeric outputs should be transcribed from the completed experiment logs; no unmeasured scores are inferred in this report.

Note: FairFace annotates race, not skin tone; results are reported per annotated race group as a limited proxy.

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GlassBox: Personalized, Privacy-Preserving, and Explainable Multimodal Risk Assessment for Online Exam Proctoring

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Abstract

The rapid expansion of online examinations has increased reliance on automated proctoring, yet fixed thresholds, isolated events, and opaque decisions can confuse legitimate individual differences with potentially unusual behavior while providing limited evidence for instructor review. GlassBox addresses this limitation by shifting proctoring from uniform surveillance toward individualized, evidence-based risk assessment. We present a browser-based framework integrating personalized behavioral calibration, multimodal temporal reasoning, interpretable machine learning, privacy-preserving processing, and empirical robustness evaluation. During an initial calibration period, GlassBox establishes an individual baseline for gaze and head-pose behavior, allowing subsequent observations to be interpreted as deviations from the student's own behavioral norm. Real-time visual signals from MediaPipe Face Landmarker are combined with browser evidence, including tab visibility, focus changes, and clipboard activity, and organized into temporal behavioral episodes capturing persistence, recurrence, and cross-signal concurrence rather than isolated events. The resulting features are processed by an interpretable logistic-regression classifier that produces probabilistic risk estimates and exposes feature-level contributions underlying each assessment. For visual robustness and fairness analysis, an open-source RetinaFace detector is evaluated independently of the real-time MediaPipe pipeline using demographically diverse facial data to quantify detection performance and subgroup disparities. The completed framework is evaluated through student- and session-independent validation, fixed-threshold and rule-based baselines, personalized-versus-global calibration comparisons, modality and component ablations, predictive-performance and probability-calibration metrics, and demographic robustness analysis. GlassBox follows a privacy-by-design, human-in-the-loop approach, performing sensitive facial processing locally wherever feasible and treating automated predictions as decision support rather than proof of misconduct. By transforming opaque, one-size-fits-all monitoring into individualized, traceable, and auditable evidence-based assessment, GlassBox provides a practical foundation for more transparent, accountable, and scientifically testable online examination proctoring.

Keywords: Online Exam Proctoring, Personalized Calibration, Explainable Al, Multimodal Risk Assessment, Privacy-Preserving Al, Algorithmic Fairness, Behavioral Risk Assessment, Human-in-the-Loop Al

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